Exponential Functions

Exam code: 9MA0

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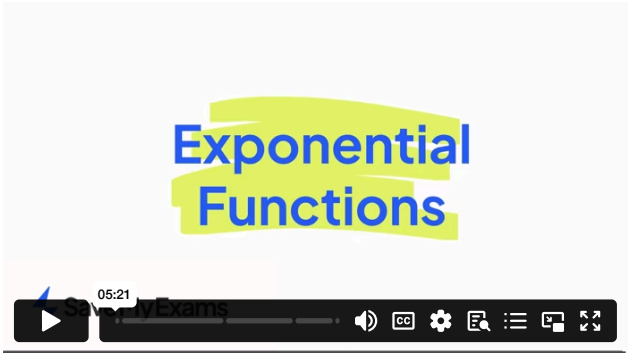

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Help me understand

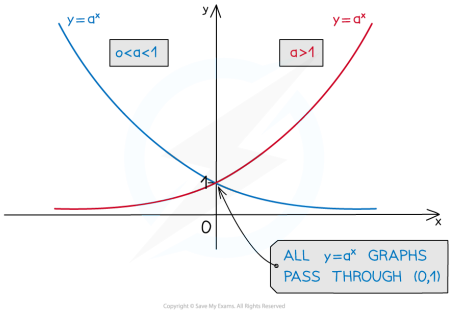
Exponential functions

What are exponential functions?

- Exponential functions of the form **y = a^x** with **a > 0** are considered at A level

What are exponential graphs?

- All graphs of the form **y = a^x** will pass through (0, 1) because **a^0 = 1**
- The **x**-axis is an **asymptote**



What is exponential growth?

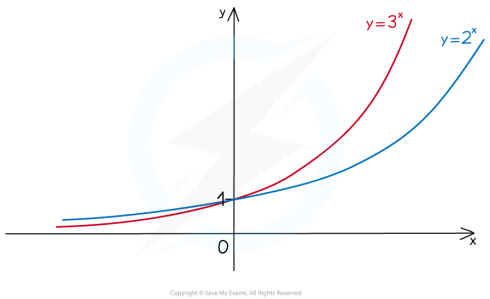
- Where **x < 0** the higher value of **a** is the "lower" graph
  - e.g. $3^x < 2^x$ when $x < 0$

Stuck on something?

Ask questions and get help with exactly what you're revising.

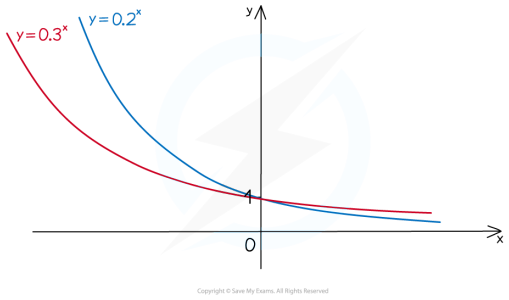
Got it

- **a > 1** is **exponential growth** (see Exponential Growth & Decay)



What is exponential decay?

- Where **x < 0** the higher value of **a** is the "lower" graph
  - e.g. $0.3^x < 0.2^x$ when $x < 0$
- Where **x > 0** the higher value of **a** is the "higher" graph
  - e.g. $0.3^x > 0.2^x$ when $x > 0$
- **0 < a < 1** is **exponential decay**

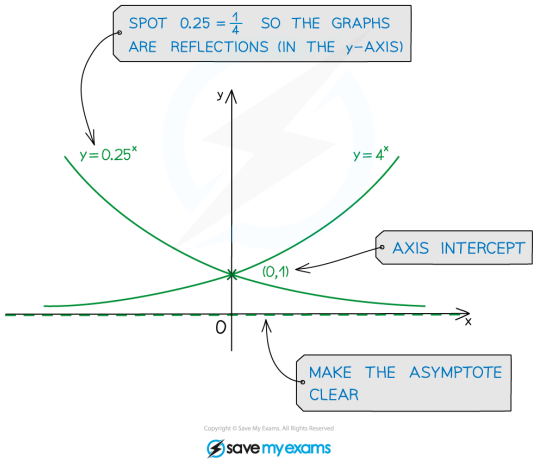


- You may like to think about why **a = 1** is **not** considered... If a = 1, y = 1^x = 1 for all values of x

Worked Example

$y = 4^x$ and $y = 0.25^x$.

Mark any points where the graphs cross the coordinate axes and make any asymptotes clear.



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